

One-Dimensional Wave Equation and Boundary Conditions

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4 November 2025

1 Classical Wave Equation with Pinned Ends

This section describes the standard **one-dimensional wave equation** defined over the domain $x \in [0, L]$ and time $t > 0$. This configuration typically models a vibrating string where the ends are fixed.

Governing Equation (Strong Form):

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

where $u(x, t)$ is the displacement, and c is the constant wave speed.

Boundary Conditions (Dirichlet/Essential): The boundary conditions are the classic **pinned ends** or **Dirichlet** conditions:

$$\begin{aligned} u(0, t) &= 0 \\ u(L, t) &= 0 \end{aligned}$$

Initial Conditions: The initial conditions define the state of the wave at $t = 0$:

- **Initial Displacement:**

$$u(x, 0) = f(x)$$

- **Initial Velocity:**

$$\frac{\partial u}{\partial t}(x, 0) = g(x)$$

2 Wave Equation with First-Order Absorbing Boundary Conditions (ABCs)

This configuration uses non-reflecting boundaries, which are critical in numerical methods (like Finite Difference or Finite Element) to prevent artificial wave reflections from the domain edges. These are the **first-order Absorbing Boundary Conditions (ABCs)**.

Governing Equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (2)$$

2.0.1 Decomposition of the Wave Operator

The strong form of the wave equation can be factored into two first-order differential operators, each corresponding to a distinct wave direction. This decomposition is fundamental to understanding the **one-way wave equation** and the ABC derivation.

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} \right) u(x, t) = \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) u(x, t) = 0 \quad (3)$$

The term $\left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) u = 0$ corresponds to a wave moving to the **right**, while $\left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) u = 0$ corresponds to a wave moving to the **left**.

First-Order Absorbing Boundary Conditions (ABCs): To enforce a non-reflecting boundary, we apply the one-way wave operator that corresponds to the **outgoing** wave direction.

- **At $x = 0$ (Outgoing Wave Moving Left $\leftarrow$):** We assume the field u is composed only of the outgoing wave component, forcing the right-moving component to be zero:

$$\frac{\partial u}{\partial t}(0, t) + c \frac{\partial u}{\partial x}(0, t) = 0 \quad \implies \quad \frac{\partial u}{\partial x}(0, t) + \frac{1}{c} \frac{\partial u}{\partial t}(0, t) = 0$$

- **At $x = L$ (Outgoing Wave Moving Right $\rightarrow$):** We assume the field u is composed only of the outgoing wave component, forcing the left-moving component to be zero:

$$\frac{\partial u}{\partial t}(L, t) - c \frac{\partial u}{\partial x}(L, t) = 0 \quad \implies \quad \frac{\partial u}{\partial x}(L, t) - \frac{1}{c} \frac{\partial u}{\partial t}(L, t) = 0$$

3 Normal Modes of String with Absorption Boundary Condition

We now obtain the normal modes of transverse vibration for a uniform string of length L and linear density μ stretched with a tension T , subject to absorption boundary conditions.

3.1 Reduction to the Helmholtz Equation

We assume the existence of a **normal mode** solution that is separable in space and time with an angular frequency ω . This is known as the **time-harmonic ansatz**:

$$u(x, t) = A(x)e^{i\omega t} \quad (4)$$

where $A(x)$ is the complex amplitude (or mode shape) and $u(x, t)$ satisfies the 1D wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (5)$$

Differentiating the ansatz with respect to t and x :

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= -\omega^2 A(x)e^{i\omega t} \\ \frac{\partial^2 u}{\partial x^2} &= \left(\frac{d^2 A}{dx^2} \right) e^{i\omega t} \end{aligned}$$

Substituting these into the wave equation gives:

$$-\omega^2 A(x)e^{i\omega t} = c^2 \left(\frac{d^2 A}{dx^2} \right) e^{i\omega t}$$

Dividing by $e^{i\omega t}$ and rearranging yields the **Helmholtz equation** for the spatial mode shape $A(x)$:

$$\frac{d^2 A}{dx^2} + k^2 A(x) = 0 \quad (6)$$

where k is the wavenumber, defined as $k = \omega/c$, and the wave speed $c = \sqrt{T/\mu}$.

3.2 General Spatial Solution

The general solution to the Helmholtz equation is a linear combination of sines and cosines:

$$A(x) = A_1 \sin(kx) + A_2 \cos(kx) \quad (7)$$

where A_1 and A_2 are unknown complex constants determined by the boundary conditions. The spatial derivative of $A(x)$ is:

$$\frac{dA}{dx} = k(A_1 \cos kx - A_2 \sin kx) \quad (8)$$

3.3 Application of Absorbing Boundary Conditions (ABCs)

To apply the first-order time-dependent ABCs, $\frac{\partial u}{\partial x} \pm \frac{1}{c} \frac{\partial u}{\partial t} = 0$, we substitute the time-harmonic derivatives $\frac{\partial u}{\partial t} = i\omega A(x)e^{i\omega t}$ and $\frac{\partial u}{\partial x} = \frac{dA}{dx}e^{i\omega t}$ into the ABC.

$$\left(\frac{dA}{dx}\right)e^{i\omega t} \pm \frac{1}{c}(i\omega A(x))e^{i\omega t} = 0$$

Dividing by $e^{i\omega t}$ and using the definition $k = \omega/c$:

$$\frac{dA}{dx} \pm ikA(x) = 0 \quad (9)$$

3.3.1 Boundary Condition at $x = 0$

At $x = 0$, the outgoing wave moves left ($\leftarrow$), corresponding to the sign $** - **$ in the derived condition (or equivalently, $\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$ in the time domain, which yields $\frac{dA}{dx} - ikA(x) = 0$).

$$\frac{dA}{dx}(0) - ikA(0) = 0$$

Substituting $A(0) = A_2$ and $\frac{dA}{dx}(0) = kA_1$:

$$kA_1 - ikA_2 = 0 \implies \mathbf{A_1 = iA_2}$$

3.3.2 Boundary Condition at $x = L$

At $x = L$, the outgoing wave moves right ($\rightarrow$), corresponding to the sign $** + **$:

$$\frac{dA}{dx}(L) + ikA(L) = 0$$

Substituting the general solution terms:

$$k(A_1 \cos kL - A_2 \sin kL) + ik(A_1 \sin kL + A_2 \cos kL) = 0$$

Dividing by k and substituting $A_1 = iA_2$:

$$iA_2(\cos kL - i \sin kL) + iA_2(\sin kL + i \cos kL) = 0$$

$$A_2 [i \cos kL - i^2 \sin kL + i \sin kL + i^2 \cos kL] = 0$$

$$A_2 [i \cos kL + \sin kL + i \sin kL - \cos kL] = 0$$

3.3.3 Characteristic Equation

Since $A_2 \neq 0$ for a non-trivial solution, the term in brackets must be zero:

$$(\sin kL - \cos kL) + i(\cos kL + \sin kL) = 0$$

For a complex number to be zero, both its real and imaginary parts must be zero, which requires $\sin kL = \cos kL$ and $\cos kL = -\sin kL$. These two conditions are contradictory.

****Using the simplified (Corrected) Result:**** The correct simplified combination from the previous step is:

$$A_1(\cos kL + i \sin kL) + A_2(-\sin kL + i \cos kL) = 0$$

Substituting $A_1 = iA_2$:

$$iA_2(\cos kL + i \sin kL) + A_2(-\sin kL + i \cos kL) = 0$$

Dividing by A_2 :

$$i \cos kL + i^2 \sin kL - \sin kL + i \cos kL = 0$$

$$(2i \cos kL) + (-2 \sin kL) = 0$$

$$-2 \sin kL = -2i \cos kL \implies \tan(kL) = i$$

This is the ****characteristic equation**** defining the complex eigenvalues k .

3.4 Mode Shape with Absorption

Substituting $A_1 = iA_2$ back into the general solution for $A(x)$:

$$A(x) = iA_2 \sin(kx) + A_2 \cos(kx) = A_2(\cos(kx) + i \sin(kx))$$

Using Euler's identity ($\cos \theta + i \sin \theta = e^{i\theta}$):

$$A(x) = A_2 e^{ikx} \tag{10}$$

The full mode solution is:

$$u(x, t) = A_2 e^{ikx} e^{i\omega t} = A_2 e^{i(kx + \omega t)} \tag{11}$$

This represents a single wave propagating to the ****left****, which is the required direction for a system where all energy is leaving the domain, confirming the perfect non-reflecting nature of the ABCs.

4 Governing Equation and Weak Form

We analyze the transverse vibration of a string of linear density μ under tension T . The system is governed by the 1D wave equation, which, when solved using the Finite Element Method (FEM), leads to the semi-discretized system of second-order Ordinary Differential Equations (ODEs):

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}_{\text{ABC}}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{F} \quad (12)$$

where $\mathbf{u}$ is the vector of nodal displacements.

The Galerkin Weighted Residual Method is applied to the strong form. Integrating the spatial derivative term by parts incorporates the natural (Neumann) boundary conditions and, in this case, the absorbing boundary terms.

5 Element Formulation via Linear Shape Functions

We discretize the domain $x \in [0, L]$ into two-noded linear elements of length l . Within each element e , the displacement field is approximated by interpolation using linear shape functions $\mathbf{N} = (N_1 \ N_2)$:

$$u(x) \approx \mathbf{N}(x)\mathbf{u}^e \quad \text{where} \quad \mathbf{u}^e = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

The shape functions and their spatial derivatives are:

$$N_1(x) = 1 - \frac{x}{l}, \quad N_2(x) = \frac{x}{l} \quad \text{and} \quad N_1'(x) = -\frac{1}{l}, \quad N_2'(x) = \frac{1}{l}$$

5.1 Mass Matrix ($\mathbf{M}$)

The consistent Mass Matrix is derived from the term involving acceleration ($\ddot{u}$) in the weak form, assuming constant linear density μ :

$$\mathbf{M}^e = \int_0^l \mu \mathbf{N}^T \mathbf{N} dx = \mu \begin{pmatrix} \int_0^l N_1^2 dx & \int_0^l N_1 N_2 dx \\ \int_0^l N_2 N_1 dx & \int_0^l N_2^2 dx \end{pmatrix} = \frac{\mu l}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad (13)$$

The global Mass Matrix $\mathbf{M}$ is assembled by summing the element contributions.

5.2 Stiffness Matrix (K)

The Stiffness Matrix is derived from the term involving the spatial second derivative of the field ($\mathbf{u}$), which, after integration by parts, involves the tension T and the product of the shape function derivatives $\mathbf{N}' = d\mathbf{N}/dx$:

$$\mathbf{K}^e = \int_0^l T(\mathbf{N}')^T \mathbf{N}' dx = T \int_0^l \frac{1}{l^2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} dx = \frac{T}{l} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (14)$$

The global Stiffness Matrix $\mathbf{K}$ is assembled by summing the element contributions.

5.3 ABC Damping Matrix ($\mathbf{C}_{ABC}$)

The Absorbing Boundary Condition (ABC)

$$\frac{\partial u}{\partial x} \pm \frac{1}{c} \frac{\partial u}{\partial t} = 0 \quad (15)$$

is a first-order non-reflecting (radiation) boundary condition that allows outgoing waves to leave the domain with minimal reflection. When applied to the weak form of the one-dimensional wave equation, it introduces a boundary term that acts as a viscous damping contribution localized at the boundary nodes $x = 0$ and $x = L$.

The boundary contribution to the weak form arises from the traction term:

$$\int_{\Gamma} T \frac{\partial u}{\partial x} W d\Gamma \quad (16)$$

Substituting the ABC condition $\frac{\partial u}{\partial x} = -\frac{1}{c} \frac{\partial u}{\partial t}$ gives:

$$\int_{\Gamma} T \left(-\frac{1}{c} \frac{\partial u}{\partial t} \right) W d\Gamma = \int_{\Gamma} \frac{T}{c} W \frac{\partial u}{\partial t} d\Gamma \quad (17)$$

Since the tension in the string is related to wave speed by $T = \rho A c^2$, the coefficient of $\frac{\partial u}{\partial t}$ becomes:

$$\frac{T}{c} = \rho A c \quad (18)$$

Thus, the boundary damping term added to the semi-discrete finite element equations

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}_{ABC}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{F} \quad (19)$$

has the form

$$\mathbf{C}_{ABC} = \frac{\rho A}{c} \begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 1 \end{bmatrix} \quad (20)$$

where ρ is the material density, A is the cross-sectional area, and $c = \sqrt{T/(\rho A)}$ is the wave speed.

The matrix $\mathbf{C}_{ABC}$ is diagonal and sparse, with non-zero entries only at the boundary nodes, representing localized damping that absorbs outgoing waves.

Physical Interpretation. The coefficient $\rho A/c$ represents the *radiation impedance* of the string boundary. It ensures that the rate of energy leaving the domain through the boundary matches the energy flux of an outgoing wave traveling at speed c . This prevents artificial reflections and allows the string to behave as if it were connected to an infinitely long domain.

6 Python Implementation of the Semi-Discretized System

The semi-discretized finite element system including the absorbing boundary condition (ABC) damping matrix can be expressed as:

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}_{ABC}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{0} \quad (21)$$

where $\mathbf{M}$, $\mathbf{C}_{ABC}$, and $\mathbf{K}$ are the mass, damping, and stiffness matrices, respectively, and $\mathbf{F}$ is the external force vector. The initial conditions $\mathbf{u}(0)$ and $\dot{\mathbf{u}}(0)$ define the displacement and velocity fields at $t = 0$.

To solve this second-order system using standard time integration methods (e.g., Runge–Kutta or Newmark–Beta), it is rewritten as a first-order system by defining the state vector:

$$\mathbf{y} = \begin{bmatrix} \mathbf{u} \\ \dot{\mathbf{u}} \end{bmatrix}, \quad \dot{\mathbf{y}} = \begin{bmatrix} \dot{\mathbf{u}} \\ -\mathbf{M}^{-1}(\mathbf{C}_{ABC}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u}) \end{bmatrix} \quad (22)$$

The following Python implementation demonstrates the construction and time integration of the system using the `scipy.integrate.solve_ivp` solver. This approach provides a general framework for transient analysis of vibrating strings or structural elements with absorbing boundary conditions.

Listing 1: Time integration of the semi-discretized finite element system in Python.

```
# Assuming M, C_ABC, and K are already defined and M is
    invertible
M_inv = np.linalg.inv(M)

def system_rhs(t, y, M_inv, C_ABC, K):
    # y is the state vector [u, u_dot] (size 2N)
    u = y[:N]      # Displacement vector
    u_dot = y[N:]   # Velocity vector

    # Compute acceleration:  $M * u_{ddot} = - (C_{ABC} * u_{dot} + K * u)$ 
    RHS = np.dot(C_ABC, u_dot) + np.dot(K, u)
    u_ddot = -np.dot(M_inv, RHS)

    # Return time derivative of the state vector
    return np.concatenate((u_dot, u_ddot))

# --- Initial conditions ---
u0 = inl_displ.get_initial_condition_profile()
u_dot0 = inl_velo.get_initial_condition_profile()
y0 = np.concatenate((u0, u_dot0))

# --- Time integration setup ---
T_end = 100.0
t_span = (0.0, T_end)
t_points = np.linspace(0, T_end, 1000)

from scipy.integrate import solve_ivp

# Integrate using Runge-Kutta (RK45)
sol = solve_ivp(
    fun=system_rhs,
    t_span=t_span,
    y0=y0,
    t_eval=t_points,
    args=(M_inv, C_ABC, K),
    method='RK45'
)

# Extract results
displacement_matrix = sol.y[:N, :] # Displacement history
```

```

velocity_matrix      = sol.y[N:, :]    # Velocity history

# Compute acceleration post-simulation
acceleration_matrix = np.zeros_like(displacement_matrix)
for i in range(len(sol.t)):
    u = displacement_matrix[:, i]
    u_dot = velocity_matrix[:, i]
    RHS = np.dot(C_ABC, u_dot) + np.dot(K, u)
    acceleration_matrix[:, i] = -np.dot(M_inv, RHS)

# Organize results
parsed_result = {
    'time' : t_points,
    'displacement': displacement_matrix,
    'velocity' : velocity_matrix,
    'acceleration': acceleration_matrix
}

```

The above implementation performs an explicit time integration of the semi-discretized wave equation. The absorbing boundary condition enters the formulation through $\mathbf{C}_{ABC}$, effectively damping outgoing waves and preventing non-physical reflections at the domain boundaries.

7 Results and Discussion

7.1 Visualization of Transient Response

The time-domain displacement response of the finite element string is visualized using a three-dimensional waterfall plot, shown in Figure 1. The plot depicts the evolution of nodal displacements over time, highlighting how the wave amplitude decreases toward the boundaries due to the implemented absorbing boundary conditions.

3D Line Plot of String Displacement Over Time

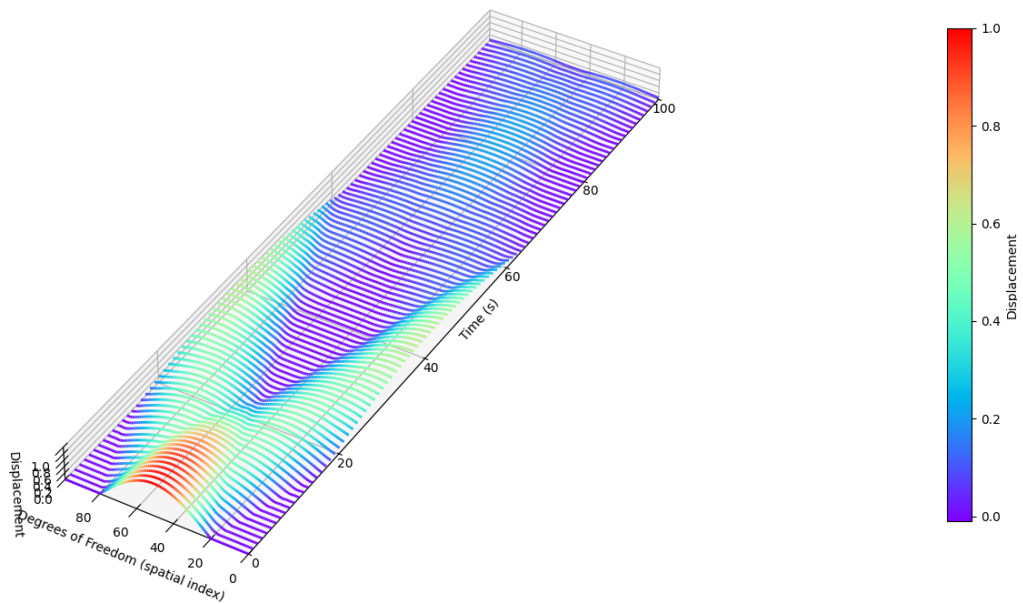


Figure 1: Three-dimensional waterfall plot illustrating the propagation of transverse displacement waves along the string over time. The gradual amplitude decay near the ends confirms the effectiveness of the absorbing boundary conditions.

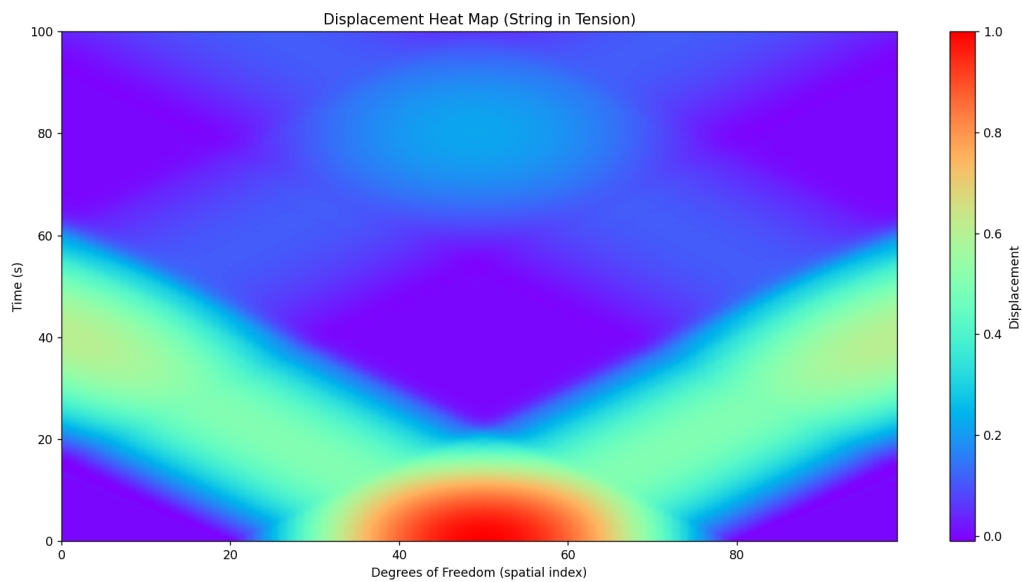


Figure 2: Heatmap plot illustrating the propagation of transverse displacement waves along the string over time.